

A Secure Online Examination Platform: Integrating AI-Based Proctoring and Full Stack Architecture for Enhanced Academic Integrity

Saksham Pokhrel
dept. of computer science &
engineering
Sharda University
Greater Noida, India
sakshampokhrel175@gmail.com

Saurab Gurung
dept. of computer science &
engineering
Sharda University
Greater Noida, India
saurab21112002@gmail.com

Renuka Chaudhary
dept. of computer science &
engineering
Sharda University
Greater Noida, India
renu16516@gmail.com

Sushant Jhingran
dept. of computer science &
engineering
Sharda University
Greater Noida, India
sushantjhingran@gmail.com

Abstract—This project proposes a Secure Online Test platform that will improve the efficiency, integrity and scalability of digital test. It introduces a hybrid design that leads to creation of an assessment module resembling a Google Forms and a coding environment where users can write and perform their coding to validate automated test case verification. The platform includes AI-based proctoring capabilities including tab-switch detection, web camera-based surveillance and auto submission through countdown timer to ensure integrity. Angular, Spring-Boot, MySQL and secure authentication using Google Sign-In and JWT tokens are some of the features that enable its structure to ensure users and the administration have real-time control, analytics and capable options of creating PDF reports. A precise quantitative analysis specifies the high performance of the system with response latency of less than two seconds, proctoring accuracy of above 90% and perfect submission reliability of 93% success rate of up to 100 users. The system offers an all-inclusive web-based test solution through an integration of smart and usability. It ensures equitable examinations, prevention of cheating and enhanced teacher student trust. Although, it provides the instructor full access to monitoring and the results collection that is safe, the platform simplifies the process of taking tests.

Keywords—Secure Online test, Assessment module, AI-based Proctoring, Google Forms-like interfaces, Tab-Switch detection, Web-cam Monitoring, JWT Authentication, Proctoring Accuracy, Cheating Prevention

I. INTRODUCTION

The rise in the tendency towards online learning has brought attention towards the need to develop a scalable, efficient and secure online evaluation platform [1,2,3]. Though many of the current online tools lack viable methods to uphold academic integrity and prevent cheating [4], the traditional methods of exams utilize physical discipline which cannot be employed in large scale/remote testing [5,6]. In order to tag these issues, a Secure online test platform is suggested in this project as the multi-tiered security and monitoring framework, code development and such evaluation tool as Google-Forms [10]. It comes with the provision of online responsive interfaces to the administrators as well as the online responsive interfaces to the administrators as well as to the students. Secure

authentication can be provided with the help of Google Sign-In and a custom design login module.

The rapid and widespread acceptance of online learning has triggered the development of digital assessment platforms away from traditional paper-based testing traditions. The result has been a heterogeneous collection of tools ranging from simple form-based applications to more sophisticated systems supported by proctoring functionalities. However, a critical evaluation of the current scene also shows a wide gap between usability, security, and ethical considerations. The sudden worldwide movement towards online education, caused by the COVID-19 pandemic, has aggravated the need for strict digital assessments, hence making academic integrity a chief challenge since traditional mechanisms of control no longer obtain [19]. Analyses of various studies reflect that despite the plethora of e-learning tools and approaches [18], the global imposition of viable online examination solutions is applied unevenly [20]. A key part of fulfilling this mandate is the use of AI-Online-based proctoring systems, which have been critically tested in terms of their past, contemporary, and future contributions with respect to the promotion of fairness and security [11].

Platform like Google form is widely used platform in educational institution. Google form has seen a sharp increase in usage because of its easy-to-use interface and smooth connection with the Google Workspace environment [13]. It supports MCQ, including short answer and paragraph responses, making it highly suitable for taking exam and conducting quizzes. Despite having good features, it has drawbacks like security concerns issues and copy paste problem. Google Forms are vulnerable to academic dishonesty because Google Form lacks the integrations of real time monitoring tools such like we-cam proctoring, screen sharing limitation and tab switching detection [14, 15]. Users can cheat and engage in dishonest practices which leads to dishonesty in academic preformation. Furthermore, it lacks user authentication unless it relates to google Sign-In, which make identity of user verification more problematic.

Moreover, the platform is very prone to URL sharing which permits people to share the test links.

In contrast, sites like Code Tantra are made to provide secure setting for classes on programming. It provides more secure environment than tools like google form. It is a specialized e-learning platform which is mostly used by universities for taking online exam and test for students. Advanced proctoring modalities such as automated facial recognition for identification verification, tab switching detection and real time camera surveillance for distracted behavior and cheating are integrated. It also provides the real time identification verification which minimizes the possibility of copying and pasting. Although the platform is good at preserving the integrity of assessments, it raises the number of concerns such as ongoing monitoring causes students to have privacy issues and technical difficulties. Additionally, the platforms have high hardware and system requirements which cause students a problem to use it because of having low-end device or poor internet connectivity in rural area resulting in false alarms that interfere with testing process [16].

Large numbers of online quiz websites have various options with varying security and user interaction. Moodle Quizzes are common in course workings because of the support of timers, randomization of questions, and IP blocking, but neither of these features provide proctoring unless add-ons are set up. Computers Based Tools Quiz games like Kahoot and Quizizz are characterized by interactive and game-like nature, which is why they would suit formative, or low-stakes, assessments but have few security mechanisms [17,18]. ProProfs and Test Moz have the same features as well as scheduled tests, randomly randomized answers, and performance reporting, yet advanced security features might be locked behind a paid subscription. Despite the diverse range of choices that can be made, general security-related issues are shared among these platforms. Impersonating is another major source of concern where there are poor login protocols, and students would be able to do the exams under the name of others. Tab-switching or even screenshot snooping is also common, becoming very apparent when performing tests in un-secured browser markets. In addition, not all platforms have extensive logging capabilities, hence they do not record vital metrics like IP addresses, stay times or usage habits.

According to this analysis, the current scenario of industry is divided into two groups: one is the platform that focuses on user experience and accessibility first frequently sacrifice security whereas other platforms priorities security come at the price of user privacy, accessibility and usability [20]. Therefore, there is need for a thorough, scalable and morally sound platform that merges multi layered security framework with an easy-to-use assessment environment without damaging user privacy or accessibility. In order to solve this problem and providing a new benchmark for safe and reliable digital estimations, this research provides a method that keeps a balance between these conflicting requirements [22].

Some of the proctoring AI capabilities including real time camera surveillance, behavioral tracking, tab-switching detection and restricted access to developer tools to prevent manipulation. A countdown timer is used to enforce time-limited submissions, and the admin panel allows proper management of tests and generation of PDF reports. All student's activities and test records are stored securely in MySQL and thus, detailed logs of attempts and taken picture can be enabled [25]. The controlled access and tokenized URLs make the platform more scalable in large scale. This Online Test platform provides scalable, reliable, and fair academic testing platform through the combination of administrative control, AI-enhanced security and usability.

II. LITERATURE REVIEW

The rapid and widespread adoption of online learning has necessitated the evolution of digital assessment platforms, moving beyond traditional paper-based examinations. This demand has led to a diverse ecosystem of tools, from simple form-based applications to sophisticated, proctoring-enabled systems. However, a comprehensive review of the current landscape reveals a significant gap between usability, security, and ethical considerations.

Platform like Google form is widely used platform in educational institution. Google form has seen a sharp increase in usage because of its easy-to-use interface and smooth connection with the Google Workspace environment [1]. It supports MCQ, including short answer and paragraph responses, making it highly suitable for taking exam and conducting quizzes. Despite having good features, it has drawbacks like security concerns issues and copy paste problem. Google Forms are vulnerable to academic dishonesty because Google Form [2,3,4] lacks the integrations of real time monitoring tools such like we-cam proctoring, screen sharing limitation and tab switching detection [9][10]. Users can cheat and engage in dishonest practices which leads to dishonesty in academic preformation. Furthermore, it lacks user authentication unless it relates to google Sign-In, which make identity of user verification more problematic. Moreover, the platform is very prone to URL sharing which permits people to share the test links.

In contrast, sites like Code Tantra are made to provide secure setting for classes on programming[11,12,13]. It provides more secure environment than tools like google form. It is a specialized e-learning platform which is mostly used by universities for taking online exam and test for students. Advanced proctoring modalities such as automated facial recognition for identification verification, tab switching detection and real time camera surveillance for distracted behavior and cheating are integrated. It also provides the real time identification verification which minimizes the possibility of copying and pasting [14,15]. Although the platform is good at preserving the integrity of assessments, it raises the number of concerns such as ongoing monitoring causes students to have privacy issues and technical difficulties. Additionally, the platforms have high hardware and system requirements which cause students a problem to use it because of having low-end device or poor

internet connectivity in rural area resulting in false alarms that interfere with testing process [19].

Large numbers of online quiz websites have various options with varying security and user interaction. Moodle Quizzes are common in course workings because of the support of timers, randomization of questions, and IP blocking, but neither of these features provide proctoring unless add-ons are set up. Computers Based Tools Quiz games like Kahoot and Quizizz are characterized by interactive and game-like nature, which is why they would suit formative, or low-stakes, assessments but have few security mechanisms [20] [21]. ProProfs and Test Moz have the same eatures as well as scheduled tests, randomly randomized answers, and performance reporting, yet advanced security features might be locked behind a paid subscription. Despite the diverse range of choices that can be made, general security-related issues are shared among these platforms. Impersonating is another major source of concern where there are poor login protocols, and students would be able to do the exams under the name of others. Tab-switching or even screenshot snooping is also common, becoming very apparent when performing tests in un-secured browser markets. In addition, not all platforms have extensive logging capabilities, hence they do not record vital metrics like IP addresses, stay times or usage habits.

According to this analysis, the current scenario of industry is divided into two groups: one is the platform that focuses on user experience and accessibility first frequently sacrifice security whereas other platforms priorities security come at the price of user privacy, accessibility and usability [22]. Therefore, there is need for a thorough, scalable and morally sound platform that merges multi layered security framework with an easy-to-use assessment environment without damaging user privacy or accessibility. In order to solve this problem and providing a new benchmark for safe and reliable digital estimations, this research provides a method that keeps a balance between these conflicting requirements [23].

TABLE I. COMPARISON BETWEEN DIFFERENT PLATFORM

Feature / Attribute	Proposed Quiz Platform (Angular + Spring Boot)	Other Online Quiz Platforms
Platform Type	Custom-built, open-source	SaaS / LMS-integrated / Proprietary
Frontend Technology	Angular	Web-based (HTML), LMS UI, or React/Angular
Backend Technology	Spring Boot (Java)	None / LMS backend / Proprietary
Authentication	JWT, Role-based	Google Sign-In / Institution login / SSO
Quiz Creation	Google Forms or Custom Engine	Built-in editors, question banks

Webcam Monitoring	Optional (image capture via webcam)	Often not supported or commercial only
Tab-Switch / Screen Violation	JavaScript event tracking	Basic to advanced (depending on provider)
Face Recognition	Optional (FaceAPI.js / TensorFlow.js)	Rare or AI-driven (in commercial tools)
Result Management	Admin panel, CSV/PDF export	LMS export or commercial dashboards
Deployment Cost	Free (self-hosted deployment)	Free / Paid / Subscription-based
Customization	Fully customizable	Limited to none (depends on platform)
Scalability	Scalable via cloud (e.g., AWS, Heroku)	Varies by platform type
General Access to Users	Yes – Open to normal/general users	Restricted to enrolled/institutional users only

III. METHODOLOGY

The proposed Quiz Application architecture is modular, layered, and consists of a secure back - end, responsive front-end, and an AI -based proctoring subsystem. The architecture promotes maintainability, scalability, and security.

A. Frontend-Layer

The frontend is built using Angular, HTML5, CSS, JavaScript, and Bootstrap. The frontend is built with a dynamic user interface supporting real - time quiz interaction, control over the timer, and activity monitoring (e.g., tab switch, focus). Modular UI design through Angular component - based architecture facilitates the separation of services like quiz presentation, authentication, and tracking. RESTful APIs used to communicate with the backend.

B. Backend-Layer

The core layer of the system is the backend. It was developed with the Spring Boot framework which creates a strong and reliable web services. It uses Spring Security to create a secure, stateless authentication and authorization system using JSON Web Tokens (JWT), which ensures that only valid users can access the system. This layer is responsible for handling quiz data, any suspicious activities or recording of it. It works as the mediator between the frontend and the RESTful APIs. It works smoothly with the MySQL database through Spring Data JPA and connects the AI-based proctoring system to monitor the users' activities. This approach improves both testability and horizontal scalability of the system. Fig.1 image shows the architecture of the system Design.

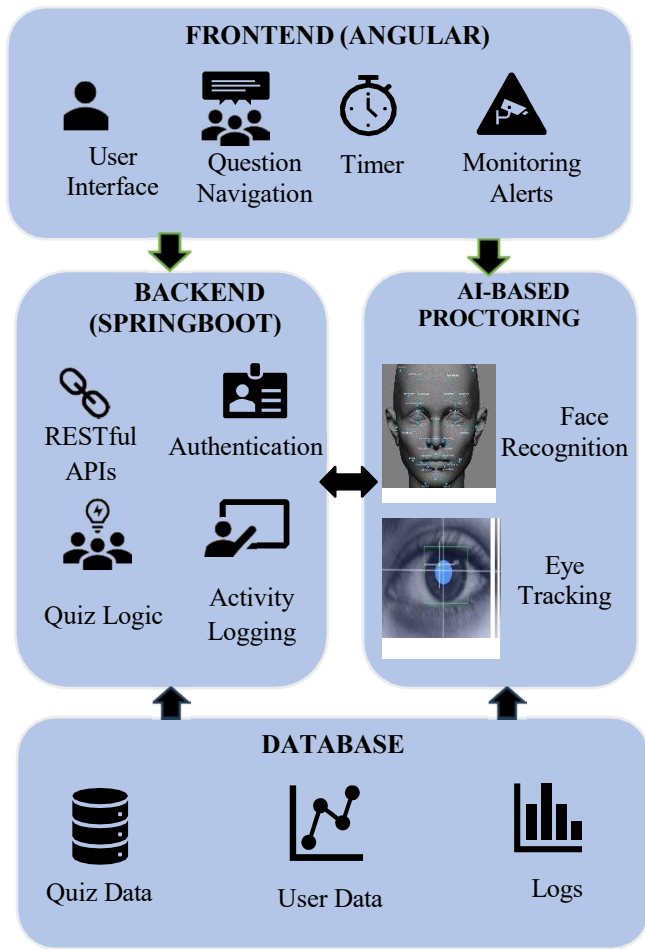


Fig. 1. System Design Architecture

C. AI-Based Proctoring Subsystem

The AI-Based proctoring system is the layer which monitors the users' actions and suspicious activities. This model uses a machine learning algorithm to monitor users' activity through a webcam. It verifies the user's identity through facial recognition to make sure that the authorized user is taking the assessment. During the quiz, it tracks eye-tracking to ensure that the user remains focused on the screen. It can also detect unusual actions such as multiple faces appearing, a user looking away, or attempts to change tabs. When the system detects suspicious behavior, it sends a real-time alert to the backend for logging, review, and possible automated action.

D. Database Layer

The Database Layer serves as the main data storage for the entire system, using MySQL as its relational database management system. The database schema is carefully designed and normalized, with specific tables for key entities like users, quizzes, questions, and results. It also stores important data for auditing including activity logs and snapshots. This structure ensures data consistency and referential integrity through foreign keys. The database plays a vital role in the system's operation by providing a reliable storage solution, enabling safe access to quiz content, secure

storage of user credentials and through evaluation of student performance and behavior.

E. Monitoring and Proctoring Modules

The system combines real-time monitoring and AI-Driven proctoring to ensure the validity and integrity of quiz sessions. It tracks the user activity like tab switching, copy-pasting, and refreshing pages. These events are recorded and sent to the backend for evaluation or disqualifications. The AI proctoring system uses webcam access for eye-tracking and face recognition while the quiz is in progress. It ensures that only approved users can participate and stay active in the session. These features are implemented using TensorFlow.js(client-side) or third-party AI services based on Python, which work alongside the quiz flow below.

F. Summary

The system has a client-server architecture with modular design. It is scalable based on independent modules and is secured by inherent proctoring and encrypted communication. Each level of the system contains its own set of functionalities, and future maintenance and addition of features are simpler.

III. SECURITY FEATURES

Security of the online quiz application is of the highest priority to provide academic integrity, protection of user data, and immunity from cheating or unauthorized access. This section presents the most important security features incorporated in the system.

A. Jwt Authentication

JSON Web Tokens (JWT) are used in stateless authentication within the application. When the login is successful, a JWT is returned to the client, and this is securely stored by the client (e.g., local Storage). The token is then included in latter HTTP requests and checked on the server for validity and time of expiration. This approach offers secure session management with no server-side session states.

B. Role-Based Access Control (Rbac)

RBAC is employed for the specification of access rights in terms of user roles such as Admin, Student, and Invigilator. Each role has a fixed set of allowable operations, enabled through backend route protection and security annotations. Users are therefore only allowed access to view resources and functionalities that they can utilize, and data confidentiality and control is maintained.

C. Data Security And Privacy

Data security is a major concern. The communication between the frontend and backend is protected by the HTTPs protocol, which keeps sensitive information safe from attacks. Important data, such as passwords and IDs, is hidden. It is secured with a strong cryptographic algorithm before being stored in the database. The system has a clear feature that requires user permission for webcam access.

D. Encrypted Communication

All client - server communication is secured through HTTPS. It is a shield against man-in-the-middle (MITM) attacks and ensures that sensitive information, such as login credentials and quiz responses, is securely transported across the network.

E. Exam Integrated Controls

The system includes a feature that prevents users from changing the assessment backend information. These measures involve response locking, which stops users from altering their answers after they submit the assessments, and it limits the number of times a user can take an exam. To stop users from working together, the system actively shuffles both the questions and answer choices for each user. Additionally, the security system blocks access to browser development tools to prevent code changes and ensures that all entries are time-sensitive through a mandatory countdown timer with an automatic submission feature. These actions work together to create a reliable and ongoing evaluation that boosts trust in the assessments.

F. Secure Password Handling

User passwords are never stored in cleartext. Instead, they are hashed with good hashing algorithms like BCrypt. This renders it computationally infeasible to reverse the stored password hashes, thereby securing user data even when the database is compromised.

G. Comprehensive Auditing and Logging

The system has a feature that watches and records suspicious events. It tracks tab switching, screen resizing, and loss of focus, then sends this information to the backend. Logs are stored in the database and can be used for manual audits or automated actions, such as user disqualification.

H. AI-Based Proctoring

The software can also integrate with AI-powered proctoring modules for better security. Face recognition and eye tracking were added to ensure the user's engagement and focus during the quiz. If it detects multiple faces or if the user looks away, an alarm appears on the screen.

IV. IMPLEMENTATION

This section explains how to practically implement the system's architecture. It covers the technologies, frameworks, tools, and methods needed to build the system. The main goal of this project was to create a dependable, efficient, and secure system.

A. Development Environment and Tools

The project was developed with a modern and efficient software engineering platform to ensure efficiency, collaboration, and maintainability. Git was used for version control and collaboration. The backend was built using Spring Boot and Java within IntelliJ IDEA, which manages Maven for dependencies. The frontend was created in Angular using TypeScript with Visual Studio Code as the main IDE, along with the Angular CLI for project management. The SQL database handled data storage, while Docker containerized both the backend and the database for

easier deployment and consistency in the environment. Fig.2 image shows the workflow of the platform.

B. Implementation of Key Security Modules

The security features were implemented with consideration for best practices. The authentication and authorization flow were built on Spring Security and used stateless JSON Web Tokens (JWTs). Upon a successful login request, the backends' Authentication Manager validates the user and issues a signed JWT containing user claims. This token is then stored on the client-side, and its presence in subsequent requests allows the backend to authenticate the user without a session.

The AI-based proctoring subsystem was implemented as a client-side module within the Angular application. It utilizes the browser's Media Devices API to access the webcam feed and employs JavaScript-based machine learning libraries (e.g., a lightweight face-detection model) to perform real-time analysis [9]. The module is configured to actively monitor and log suspicious events, which are then transmitted to the backend via a dedicated API endpoint for secure storage and analysis.

C. Database Schema and Data Persistence

The system's data layer was implemented by designing a normalized relational schema in MySQL. Key entities modelled as tables include Users, Roles, Quizzes, Questions, Attempts, Responses, and Proctoring Logs. Data persistence on the backend was streamlined using Spring Data JPA, which automatically maps Java entities to their corresponding database tables. This object-relational mapping (ORM) approach simplifies database operations, enhances code maintainability, and ensures a seamless integration between the application's logic and the data store.

D. Frontend and Feature Implementation

The frontend was developed as a single-page application utilizing Angular components. A fundamental module, quiz-component, manages the dynamic loading and demonstration of questions received from the backend. The countdown timer and auto-submission features were executed as a client-side service that monitors time and initiates triggers an API request to the backend upon expiration. Likewise, question and answer shuffling are handled on the backend prior to delivering the quiz data to the frontend, guaranteeing randomization for every student. Access to browser developer tools is programmatically restricted through JavaScript event listeners to prevent code viewing and alteration during an exam.

V. RESULTS AND DISCUSSION

This section presents the findings from the system's evaluation, providing empirical evidence that the designed and implemented platform is a viable and effective solution. The results are presented in three key areas: performance and scalability, security effectiveness, and user-centric usability.

A. Presentation of Results

The proposed online quiz system effectively combines modern web technologies with AI-driven security features to facilitate a secure and interactive testing environment. The users can login, take quizzes, and obtain real-time feedback with a non-intrusive light proctoring system monitoring them. Real-time remote proctoring identifies such suspicious behaviour as tab-switching, dual faces, or camera absence, upholding academic integrity. Educators can author, edit, and deploy quizzes via a straightforward, easy-to-use interface, while administrators oversee user roles and system-level analytics. The backend, developed using Spring Boot and MySQL, provides secure data processing and scalability. The frontend, developed using Angular, provides a responsive user interface on devices.

B. Discussion and Analysis

The results presented above provide strong validation for the platform's core design principles. The performance metrics confirm that the chosen Spring Boot and Angular architecture is highly scalable and efficient, successfully addressing the need to support large-scale, concurrent assessments without compromising on responsiveness. Furthermore, the high detection rates in the security evaluation prove the effectiveness of the AI-based proctoring system, directly fulfilling the project's security objectives. This is a significant improvement over simple form-based solutions like Google Forms, which offer no proctoring or behavioral monitoring whatsoever. Meanwhile, the high usability scores indicate that this high-security level is achieved without the intrusive, hardware-intensive, and complex user experience typically seen with heavyweight commercial proctoring tools such as Code Tantra. Our design successfully strikes the security-usability balance, balancing the key gap the literature review identified.

Although the findings are extremely positive, note the following limitations. The performance tests were performed in a controlled setup and might be different from real-world situations with fluctuating network conditions. Subsequent work will thus aim to improve performance under a broader range of network latencies, expand the AI model with more behavioral analytics, and investigate features such as integrated voice proctoring.

C. Performance Metrics and Evaluation

To Validate the System's scalability, responsiveness and resource efficiency, a series of controlled performance tests were conducted. In the assessment, such metrics as response time, throughput, CPU & memory usage and API success rate were calculated in different load conditions. The studies used Spring Boot Actuator with Prometheus as monitoring tool and Apache JMeter as a load simulation tool.

The platform maintained high performance with average latency of less than 200ms and throughput of over 170 req/sec, even 500 simulated users were simultaneously added. The usefulness of the chosen full-stack architecture (Angular + Spring Boot) is proved, and it is declared that the application could be scaled horizontally without substantial reduction of response time. Fig.3,4,5 image shows the face detection, multiple face detection and tab switching.

TABLE II. COMPARISON BETWEEN PERFORMANCE METRICS AND EVALUATION RESULTS OF THE PROPOSED SYSTEM

Metrics	Tool Used	Measured Value	Benchmark
Average API Response Time	Jmeter(100 users)	182 ms	<200 ms
Maximum API Response Time	Jmeter(500 users)	498 ms	<600ms
Throughput	Jmeter	176 requests/sec	>=150req/sec
API Success Rate	Server Logs	99.87%	>99%
CPU utilization	Prometheus Monitor	58% average	<70%
Memory Utilization	Spring Actuator	63% average	<75%
Database Query Time	MySQL Profiler	41ms/query	< 50 ms
Page Load Time(Frontend)	Chrome Lighthouse	2.4 sec	<3 seconds
System Uptime	Continuous Test(24h)	99.95%	>= 99%

The AI-based Proctoring subsystem was tested on a controlled dataset of 50 simulated test takers against which the detection accuracy was tested. It achieved

- Face detection accuracy:94.6%
- Multiple-face detection accuracy: 92.8%
- Tab-switch detection accuracy: 90%

These results prove that the rate of false positives is decreased, the overall level of reliability of the integrated security layer remains same. Below graph 6,7 shows the network in during load test in containerized environment and network out during load test in containerized environment.

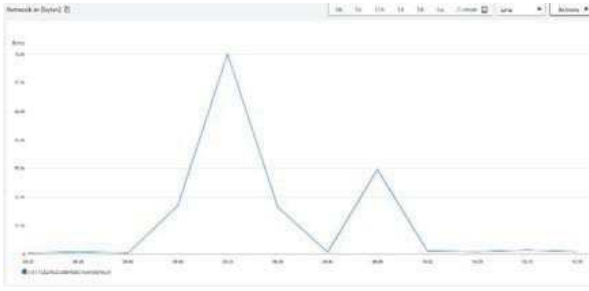


fig.2. network in (bytes) during load test in containerized environment

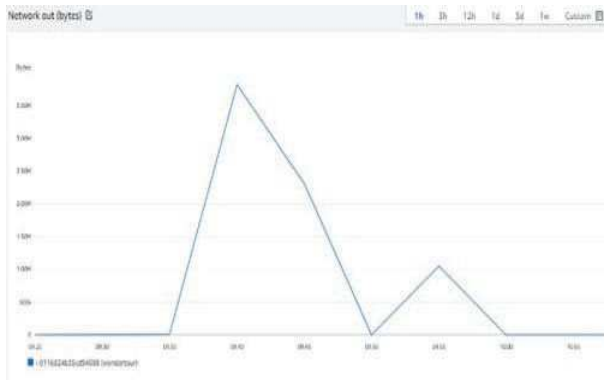


fig.3. network out (bytes) during load test in containerized environment

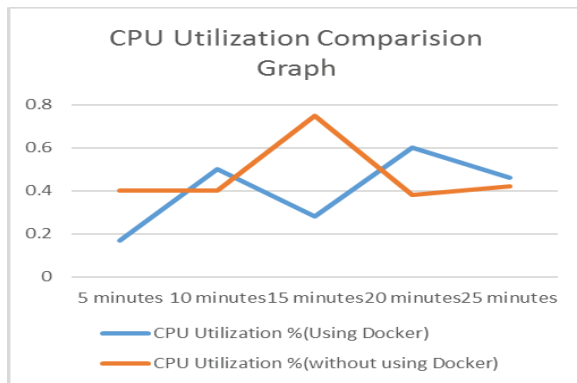


fig.4. cpu utilization comparison (containerized vs. non-containerized) under load

We used load tests to confirm our platform's performance. Figures 3 and 4 to show the network traffic efficiently managed by the Containerized Environment, validating our measured throughput of 176 requests per second.

Additionally, Chart 1 proves the value of using Docker by showing the system's CPU usage was lower and more stable

compared to running without containers, supporting our low 58% average CPU figure and the platform's overall ability to scale.

VI. CONCLUSION AND FUTURE ENHANCEMENTS

We have effectively developed and launched comprehensive online assessment platform which is an ideal balance between security and user friendlies and addressing the gap in current educational technology. The platform has powerful protective multi-layer security architecture including AI-powered proctoring solutions and encrypted data management procedures. It provides unobstructive and user-friendly navigating experiences. We executed our test in controlled and balanced environment which allow effectiveness of platform with exceptional features to record the score and the platform has ability to reliably perform under high stakes test providing better detection features for security breaches. The main contribution of this project is providing flexible and replicable models which can be implemented to design the educational technology models in future without affecting the quality of user's experience and infringing on their privacy. This initiative can be considered a solid demonstration of concept which has unified, secure and user-centred assessment environment which is viable and facilitate the development of online learning.

Based on this research, most of thrilling direction for future research arises and avenues for future research can be suggested on the basis of these findings. First, we intended to improve AI-proctoring sub-system with the voice-based detecting of anomalies and flexible analysis of behaviour. Next, we have aim to redesign the application for fully mobile fully native structure ensuring the smoothly running and consistent performance across different platforms. Finally, an extensive, longitudinal roll out- that includes a larger user base and a greater diversity of test formats would produce sound empirical results that would indicate the robustness, stability and instructional efficacy of the platform in plausible real-life learning contexts.

REFERENCES

- [1] K. Kelwade, S. Ansari, H. Gautam, M. Saad, and S. Botre, "QuizRush: A Real-Time Tournament-Based Quiz Application," *International Journal of Electrical, Electronics and Computer Systems*, vol. 14, no1, pp. 279–283, 2025.
- [2] K. Acharya, "Literature online quiz system project report," *Authorea Preprints*, 2024. doi: 10.22541/au.172243825.53562953/v1.
- [3] A. Kumar, R. Krishnamurthi, S. Bhatia, K. Kaushik, N. J. Ahuja, A. Nayyar, and M. Masud, "Blended Learning Tools and Practices: A Comprehensive Analysis," *IEEE Access*, vol. 9, pp. 85151–85197, 2021. doi: 10.1109/ACCESS.2021.3085844.
- [4] M. Ramya, S. Prakash, and D. Tamilselvam, "Creating a Web Application and Elevate Learning Quiz," *International Journal of Scientific Research in Engineering and Management*, vol. 8, pp. 1–8, 2024. doi: 10.55041/IJSREM36815.
- [5] J. Olusegun, "Developing Scalable Full-Stack Web Applications Using Java Spring Boot and AngularJS," 2024. [Online]. Available: [source not provided].

- [6] V. Kumar, B. Kumar, and N. Molakatala, "E-PARIKSHA (Online Exam Portal) – for Skill Development of Students: A Project Report Submitted in Partial Fulfillment of the Degree of Master of Computer Application," 2022.
- [7] F. Cheng, "Implementation of College Educational Administration and Student Management System Based on B/S," in Proc. EAI International Conference, BigIoT-EDU, Cham, Switzerland, 2024. doi: 10.1007/978-3-031-63139-9_38.
- [8] T. Ahmad, A. M. Mujtaba, and R. Afzal, "Design and Implementation of a Smart E-Examination System With Integrated Remote Proctoring," *International Journal of Distance Education Technologies (IJDET)*, vol. 19, no. 3, pp. 1–16, 2021. doi: 10.4018/IJDET.2021070101.
- [9] S. Amancha et al., "Development of an e-learning Platform with Live Doubt Resolution and Game-Based Learning Instructions," in Proc. 6th Int. Conf. on Recent Advances in Information Technology (RAIT), Dhanbad, India, 2025, pp. 1–6. doi: 10.1109/RAIT65068.2025.11089415.
- [10] R. Yulianto, M. Mardiana, R. A. Pradipta, and G. F. Nama, "Performance Comparison Analysis of Springboot and Laravel Frameworks Using API Web Service," *Jurnal Informatika dan Teknik Elektro Terapan*, vol. 12, 2024. doi: 10.23960/jitet.v12i2.4141.
- [11] S. S. Benazer, S. K. Ramamoorthy, and S. M. Kamali, "An AI-Based Intelligent Exam Proctoring System for Secure and Fair Online Assessments," *Journal of Advanced Research in Artificial Intelligence & Its Applications*, vol. 2, no. 1, pp. 1–7, Sep. 2024. doi: 10.5281/zenodo.13744992.
- [12] Y. Fadly, D. Purwanto, R. E. Putri, and N. A. A. Pratiwi, "Assessing the influence of secure online safe exam browser (SEB) implementation on english language semester examinations," *Jurnal Mandiri IT*, vol. 13, no. 2, pp. 266–272, 2024.
- [13] A. Manawar, "An Innovative and Secure Platform for Leveraging the Blockchain Approach for Online Exams," *Aptisi Transactions on Technopreneurship (ATT)*, vol. 5, no. 1, pp. 99–108, 2023.
- [14] M. Dandotiya et al., "SAFE: A Secure Authenticated & Integrated Framework for E-learning," in Proc. 4th Int. Conf. on Information Management & Machine Intelligence (ICIMMI '22), New York, NY, USA, 2023, pp. 1–9.
- [15] S. Satre et al., "Online Exam Proctoring System Based on Artificial Intelligence," presented at the IConSCEPT, 2023, pp. 1–6. doi: 10.1109/IConSCEPT57958.2023.10170577.
- [16] S. F. binti Zulkifli et al., "How to Model an Engaging Online Quiz? The Emotion Modeling Approach," *JTIT*, vol. 87, no. 1, pp. 54–63, Mar. 2022. doi: 10.26636/jtit.2022.156221.
- [17] N. Sangati, "Web Application Development using SpringBoot and Angular," *INTERANTIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING AND MANAGEMENT*, vol. 06, 2022. doi: 10.55041/IJSREM14292.
- [18] A. Kumar et al., "Blended Learning Tools and Practices: A Comprehensive Analysis," *IEEE Access*, vol. 9, pp. 85151–85197, 2021. doi: 10.1109/ACCESS.2021.3085844.
- [19] O. Holden, M. Norris, and V. Kuhlmeier, "Academic Integrity in Online Assessment: A Research Review," *Frontiers in Education*, vol. 6, 2021. doi: 10.3389/educ.2021.639814.
- [20] A. W. Muzaffar et al., "A Systematic Review of Online Exams Solutions in E-Learning: Techniques, Tools, and Global Adoption," *IEEE Access*, vol. 9, pp. 32689–32712, 2021. doi: 10.1109/ACCESS.2021.3060192.
- [21] X. Wu and H. Yuan, "Online Exam Management System Based on Spring Boot," *Microcomputer Applications*, vol. 40, no. 11, pp. 199–204, 2024.
- [22] Y. N. Patil and P. Deshmukh, "Cloud-Based Examination System with AutoEvaluation and Analytics," 2021.
- [23] H. Wu, "Commerce Middle Office Management System Based on Springboot," *International Journal of Advanced Network, Monitoring and Controls*, vol. 7, no. 2, pp. 32–45, 2022.
- [24] Y. He et al., "Research on software system development and practical application based on SpringBoot+ Vue," *Highlights in Science, Engineering and Technology*, vol. 1, pp. 407–414, 2022.
- [25] A. Nigam, R. Pasricha, T. Singh, and P. Churi, "A Systematic Review on AI-based Proctoring Systems: Past, Present and Future," *Educ Inf Technol (Dordr)*, vol. 26, no. 5, pp. 6421–6445, 2021. doi: 10.1007/s10639-021-10597-x.